

LINEAR ALGEBRA FOR ENGINEERS: ASSIGNMENT PROBLEMS

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1. VECTOR SPACE

1.1. Assignment-Problems: Vector spaces. In the following a set V , a field $\mathbb{F}$, which is either $\mathbb{R}$ or $\mathbb{C}$, and operations of addition $+$ and scalar multiplication $\cdot$, are given. For $\alpha \in \mathbb{F}$ and $x \in V$, we write their multiplication $\alpha \cdot x$ as αx . Check whether V is a vector space over $\mathbb{F}$, with these operations.

- (1) $V = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 = 0\}$, $\mathbb{F} = \mathbb{R}$, with $+$ and $\cdot$ as in $\mathbb{R}^2$.
- (2) $V = \{(x_1, x_2) \in \mathbb{R}^2 : 2x_1 + 3x_2 = 0\}$, $\mathbb{F} = \mathbb{R}$, with $+$ and $\cdot$ as in $\mathbb{R}^2$.
- (3) $V = \{x = (x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2 = 1\}$, $\mathbb{F} = \mathbb{R}$, with $+$ and $\cdot$ as in $\mathbb{R}^2$.
- (4) $V = \mathbb{R}^2$, $\mathbb{F} = \mathbb{R}$ with $+$ as in $\mathbb{R}^2$ and for $x = (x_1, x_2) \in V$ and $\alpha \in \mathbb{R}$,
 $\alpha x := (x_2, \alpha x_1)$.
- (5) $V = \mathbb{C}^2$, $\mathbb{F} = \mathbb{C}$. For $x = (x_1, x_2)$, $y = (y_1, y_2)$ and $\alpha \in \mathbb{C}$,

$$x + y := (x_1 + 2y_1, x_2 + 3y_2), \quad \alpha x := (\alpha x_1, \alpha x_2).$$

- (6) $V = \mathbb{R}^2$, $\mathbb{F} = \mathbb{R}$ with $+$ as in $\mathbb{R}^2$ and for $x = (x_1, x_2) \in V$ and $\alpha \in \mathbb{R}$,
 $\alpha x := (x_1, 0)$.
- (7) $V = \mathbb{R}^2$, $\mathbb{F} = \mathbb{R}$ with $+$ as in $\mathbb{R}^2$ and for $x = (x_1, x_2) \in V$ and $\alpha \in \mathbb{R}$,
 $\alpha x := (\alpha x_1, -\alpha x_2)$.
- (8) $V = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 x_2 = 0\}$, $\mathbb{F} = \mathbb{R}$, with $+$ and $\cdot$ as in $\mathbb{R}^2$.
- (9) $V = \{x \in \mathbb{R} : x \geq 0\}$, the set of non-negative real numbers, and $\mathbb{F} = \mathbb{R}$. For
 $x, y \in V$, $\alpha \in \mathbb{R}$,

$$x + y := xy, \quad \alpha x := |\alpha|x.$$

- (10) $V = \mathcal{R}([a, b], \mathbb{R})$, the set of all real valued Riemann integrable functions on $[a, b]$, and $\mathbb{F} = \mathbb{R}$, with usual addition and scalar multiplication of functions.
- (11) V is the set of all polynomials of degree 5 with real coefficients and $\mathbb{F} = \mathbb{R}$, with usual addition and scalar multiplication of polynomials.
- (12) Let S be a non-empty set and $s_0 \in S$. Let V be the set of all functions $f : S \rightarrow \mathbb{R}$ with $f(s_0) = 0$ and $\mathbb{F} = \mathbb{R}$, with usual addition and scalar multiplication of functions.
- (13) V is the set of all functions $f : \mathbb{R} \rightarrow \mathbb{C}$ satisfying $f(-t) = \overline{f(t)}$ and $\mathbb{F} = \mathbb{R}$, with usual addition and scalar multiplication of functions.

1.2. Assignment-Problems: Subspaces.

- (1) In the following, vector space V over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ is given with usual operations on it. Check whether the given subset U is a subspace of V .
 - (a) $V = \mathcal{P}_3$ and $U = \{a_1t + a_2t^2 + a_3t^3 : a_0 = 0\}$.
 - (b) $V = \mathcal{P}_3$ and $U = \{a_0 + a_1t + a_2t^2 + a_3t^3 : a_2 = 0\}$.
 - (c) $V = \mathcal{P}_3$ and $U = \{a_0 + a_1t + a_2t^2 + a_3t^3 : a_1 + a_2 + a_3 + a_4 = 0\}$.
 - (d) V is the plane $\mathbb{R}^2$ and U is any straight line passing through the origin.
 - (e) $V = \mathbb{R}^2$ and $U = \{(x_1, x_2) : x_2 = 2x_1 - 1\}$.
 - (f) $V = \mathbb{R}^3$ and $U = \{(x_1, x_2, x_3) : 2x_1 - x_2 - x_3 = 0\}$.
 - (g) $V = \mathcal{C}[-1, 1]$ and $U = \{f \in V : f \text{ is an odd function}\}$.
 - (h) $V = \mathcal{C}[0, 1]$ and $U = \{f \in V : f(t) \geq 0 \text{ for all } t \in [0, 1]\}$.
 - (i) $V = \mathcal{C}^k[a, b]$ and $U = \mathcal{P}[a, b]$ for each $k \geq 1$.
 - (j) $V = \mathcal{C}[0, 1]$ and $U = \{f \in V : f \text{ is differentiable}\}$.
 - (k) $V = \mathcal{C}[-1, 1]$ and $U = \{f \in V : f \text{ is an odd function}\}$.
 - (l) $V = \mathcal{P}_3(\mathbb{R})$ and $U = \{p \in V : p(0) = 0\}$.
 - (m) $V = \mathcal{P}_3(\mathbb{C})$ and $U = \{p \in V : p(1) = 0\}$.
 - (n) $V = \mathcal{P}_3(\mathbb{C})$ and $U = \{a + bt + ct^2 + dt^3 : a, b, c, d \text{ integers}\}$.
- (2) For $\alpha \in \mathbb{F}$, let $V_\alpha = \{(a, b, c) \in \mathbb{F}^3 : a + b + c = \alpha\}$. Show that V_α is a subspace of $\mathbb{F}^3$ iff $\alpha = 0$.
- (3) Give an example of a nonempty subset of $\mathbb{R}^2$ which is closed under scalar multiplication but is not a subspace of $\mathbb{R}^2$.
- (4) Suppose U is a subspace of V and V is a subspace of W . Show that U is a subspace of W .
- (5) Give an example of two subspaces of $\mathbb{C}^3$ whose union is not a subspace of $\mathbb{C}^3$.
- (6) Show by a counter-example that if $U + W = U + X$ for subspaces U, W, X of V , then W need not be equal to X .
- (7) Let $m \in \mathbb{N}$. Does the set $\{0\} \cup \{p \in \mathcal{P} : \text{degree of } p \text{ is equal to } m\}$ form a subspace of $\mathcal{P}$?
- (8) Let $u_1, \dots, u_n$ be vectors in a vector space V . Let $U := \{\alpha_1 u_1 + \dots + \alpha_n u_n : \alpha_1, \dots, \alpha_n \in \mathbb{F}\}$. Show that U is a subspace of V .
- (9) Prove that the only non-trivial proper subspaces of $\mathbb{R}^2$ are straight lines passing through the origin.

1.3. Assignment-Problems: Linear span.

- (1) Let $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$ and $e_3 = (0, 0, 1)$. Show that $\text{span}\{e_1 + e_2, e_2 + e_3, e_3 + e_1\} = \mathbb{R}^3$
- (2) What is $\text{span}\{t^n : n = 0, 2, 4, 6, \dots\}$?
- (3) Do the polynomials $t^3 - 2t^2 + 1$, $4t^2 - t + 3$, and $3t - 2$ span $\mathcal{P}_3$?
- (4) Let $u_1(t) = 1$, and for $j = 2, 3, \dots$, let $u_j(t) = 1 + t + \dots + t^j$. Show that $\text{span}\{u_1, \dots, u_n\}$ is $\mathcal{P}_n$, and $\text{span}\{u_1, u_2, \dots\}$ is $\mathcal{P}$.
- (5) We know that $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$. Does $\sin x \in \text{span}\{1, x, x^2, x^3, \dots\}$?
- (6) Let $U = \{(a, b, c, d) \in \mathbb{R}^4 : 5a + 2b - c = 3a + 2c - d = 0\}$. Find a finite subset S of U such that $\text{span}(S) = U$.
- (7) Let x, y, z be vectors in a vector space V . Show that if $x + y + z = 0$, then $\text{span}\{x, y\} = \text{span}\{x, z\} = \text{span}\{y, z\}$.
- (8) Let S be a subset of a vector space V . Show that S is a subspace of V iff $S = \text{span}(S)$.
- (9) Let U be a subspace of a vector space V . Let $w \in V \setminus U$. Show that for every $x \in \text{span}(U \cup \{w\})$, a unique pair $(\alpha, u) \in \mathbb{F} \times U$ exists such that $x = u + \alpha w$.
- (10) Let V be a vector space; $x, y \in V$; U is a subspace of V ; $X = \text{span}(U \cup \{x\})$; $Y = \text{span}(U \cup \{y\})$. Suppose that $y \in X \setminus U$. Show that $x \in Y$.
- (11) Let $u, v, w_1, \dots, w_n$ be distinct vectors in a vector space V ; $B = \{u, w_1, \dots, w_n\}$; and $E = \{v, w_1, \dots, w_n\}$. Prove that $\text{span}(B) = \text{span}(E)$ iff $u \in \text{span}(B)$ iff $v \in \text{span}(E)$.
- (12) Let S be a proper subset of a vector space. Show that if x is a vector such that $x \in \text{span}(S \setminus \{x\})$, then $\text{span}(S \setminus \{x\}) = \text{span}(S)$.
- (13) Let A and B be subsets of a vector space V . Show that $\text{span}(A \cap B) \subseteq \text{span}(A) \cap \text{span}(B)$. Give an example to show that $\text{span}(A) \cap \text{span}(B)$ need not be a subset of $\text{span}(A \cap B)$.
- (14) Let $\mathcal{P}_e = \{p(t) \in \mathcal{P} : p(-t) = p(t)\}$; $\mathcal{P}_o = \{p(t) \in \mathcal{P} : p(-t) = -p(t)\}$. Show that both $\mathcal{P}_e$ and $\mathcal{P}_o$ are subspaces of $\mathcal{P}$ and that $\mathcal{P} = \mathcal{P}_e \oplus \mathcal{P}_o$.
- (15) Let $U = \{0, \dots, 0, a_{n+1}, \dots, a_{2n} : a_i \in \mathbb{R}\}$; $V = \{(a_1, \dots, a_{2n}) \in \mathbb{R}^{2n} : a_i = a_{n+i} \text{ for } i = 1, \dots, n\}$. Does it follow that $\mathbb{R}^{2n} = U \oplus V$?
- (16) Construct three subspaces U, W, X of a vector space V so that $V = U \oplus W$ and $V = U \oplus X$ but $W \neq X$.
- (17) Suppose U and W are subspaces of a vector space V . Show that $U + W = U$ iff $W \subseteq U$.

1.4. Assignment-Problems: Linear independence.

- (1) Is $\{(1, 1, 0, 2), (1, 1, 3, 2), (4, 2, 1, 2)\}$ linearly independent in $\mathbb{R}^4$?
- (2) Is $\{(6, 2, 1), (4, 3, -1), (2, 4, 1)\}$ linearly independent in $\mathbb{C}^3$?
- (3) Let $\{u, v, w, x\}$ be linearly independent in a vector space V . Does it imply that $\{u + v, v + w, w + x, x + u\}$ is linearly independent in V ?
- (4) Show that the vectors $(1, 0, 0), (0, 2, 0), (0, 0, 3)$ and $(1, 2, 3)$ are linearly dependent, but any three of them are linearly independent.
- (5) Give three vectors in $\mathbb{R}^2$ such that none of the three is a scalar multiple of another.
- (6) In each of the following, a vector space V and $S \subseteq V$ are given. Determine whether S is linearly dependent and if it is, express one of the vectors in S as a linear combination of some or all of the remaining vectors.
 - (a) $V = \mathbb{R}^3$, $S = \{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$.
 - (b) $V = \mathbb{R}^3$, $S = \{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$.
 - (c) $V = \mathbb{R}^3$, $S = \{(1, -3, -2), (-3, 1, 3), (2, 5, 7)\}$.
 - (d) $V = \mathbb{P}^3$, $S = \{t^2 - 3t + 5, t^3 + 2t^2 - t + 1, t^3 + 3t^2 - 1\}$.
 - (e) $V = \mathbb{P}^3$, $S = \{-2t^3 - 11t^2 + 3t + 2, t^3 - 2t^2 + 3t + 1, 2t^3 + t^2 + 3t - 2\}$.
 - (f) $V = \mathbb{P}^3$, $S = \{6t^3 - 3t^2 + t + 2, t^3 - t^2 + 2t + 3, 2t^3 + t^2 - 3t + 1\}$.
 - (g) V is the set of all real valued functions defined on $\mathbb{R}$ and $S = \{2, \sin^2 t, \cos^2 t\}$.
 - (h) V as above and $S = \{1, \sin t, \sin 2t\}$.
 - (i) $V = C([-\pi, \pi])$, $S = \{\sin t, \sin 2t, \dots, \sin nt\}$ for some $n \in \mathbb{N}$.
- (7) Let $p_1(t) = 1 + t + 3t^2$, $p_2(t) = 2 + 4t + t^2$, $p_3(t) = 2t + 5t^2$. Are the polynomials p_1, p_2, p_3 linearly independent?
- (8) Prove that in the vector space of all real valued functions defined on $\mathbb{R}$, the set $\{e^x, xe^x, x^3e^x\}$ of functions is linearly independent.
- (9) Let $V = \mathbb{C}^2$. Determine conditions on $\alpha, \beta \in \mathbb{C}$ such that the vectors $(\alpha, 1), (\beta, 1)$ in $\mathbb{C}^2$ are linearly dependent.
- (10) Let $V = \mathbb{C}^3$. Determine conditions on $\alpha, \beta, \gamma \in \mathbb{C}$ such that the vectors $(1, \alpha, \alpha^2), (1, \beta, \beta^2)$ and $(1, \gamma, \gamma^2)$ in $\mathbb{C}^3$ are linearly dependent.
- (11) Let E be any subset of a vector space V . Prove that E is linearly dependent iff there exist finite number of vectors in E which are linearly dependent.
- (12) Let E be any subset of a vector space V . Prove that E is linearly independent iff there exist finite number of vectors in E which are linearly independent.
- (13) Prove that 0 does not belong to any linearly independent set.
- (14) Prove that all supersets of a linearly dependent sets are linearly dependent.
- (15) Prove that all subsets of a linearly independent set are linearly independent.
- (16) Let U be a subspace of V . Let E be a linearly independent set in U . Prove that E is a linearly independent in V .
- (17) Suppose S is a set of vectors and some $v \in S$ is not a linear combination of other vectors in S . Does it follow that S linearly independent?
- (18) Show that two vectors (a, b) and (c, d) in $\mathbb{R}^2$ are linearly independent if and only if $ad - bc \neq 0$.

1.5. Assignment-Problems: Basis, Dimension and Quotient space.

- (1) Is $\{-1 - t - 2t^2, 2 + t - 2t^2, 1 - 2t + 4t^2\}$ a basis for $\mathcal{P}_2$?
- (2) Is $\{1 + 2t + t^2, 3 + t^2, t + t^2\}$ a basis for $\mathcal{P}_2$?
- (3) Is $\{1 + 2t + 3t^2, 4 - 5t + 6t^2, 3t + t^2\}$ a basis for $\mathcal{P}_2$?
- (4) Let $\{x, y, z\}$ be a basis for a vector space V . Check whether $\{x + y, y + z, z + x\}$ is a basis for V ?
- (5) Extend the set $\{1 + t^2, 1 - t^2\}$ to a basis of $\mathcal{P}_3$.
- (6) Find a basis for the subspace $\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$ of $\mathbb{R}^3$.
- (7) Construct a basis for $\{(x_1, \dots, x_6) \in \mathbb{R}^6 : x_2 = 2x_1, x_4 = 4x_3, x_6 = 6x_5\}$.
- (8) Does there exist a basis for $\mathcal{P}^4$ none of the vectors of which is of degree 3?
- (9) Under what conditions on α , $\{(1, \alpha, 0), (\alpha, 0, 1), (1 + \alpha, \alpha, 1)\}$ is a basis of $\mathbb{R}^3$?
- (10) Is $\{1 + t^n, t + t^n, \dots, t^{n-1} + t^n, t^n\}$ a basis for $\mathcal{P}_n$?
- (11) Let $u_1 = 1$ and let $u_j = 1 + t + t^2 + \dots + t^{j-1}$ for $j \geq 2$. Prove or disprove: $\text{span}\{u_1, \dots, u_n\} = \mathcal{P}_n$ and $\text{span}\{u_1, u_2, \dots\} = \mathcal{P}$.
- (12) Let $v_1, \dots, v_n$ be linearly independent vectors in a vector space V . Suppose $w \in V$ is such that the vectors $w + v_1, \dots, w + v_n$ are linearly dependent. Show that $w \in \text{span}\{v_1, \dots, v_n\}$.
- (13) Find bases and dimensions of the following subspaces of $\mathbb{R}^5$:
 - (a) $\{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 : x_1 - x_3 - x_4 = 0\}$.
 - (b) $\{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 : x_2 = x_3 = x_4, x_1 + x_5 = 0\}$.
 - (c) $\text{span}\{(1, -1, 0, 2, 1), (2, 1, -2, 0, 0), (0, -3, 2, 4, 2), (3, 3, -4, -2, -1), (2, 4, 1, 0, 1), (5, 7, -3, -2, 0)\}$.
- (14) Find $\dim(\text{span}\{1 + t^2, -1 + t + t^2, -6 + 3t, 1 + t^2 + t^3, t^3\})$.
- (15) Find a basis, and hence dimension, for each of the following subspaces of the vector space V of all twice differentiable functions from $\mathbb{R}$ to $\mathbb{R}$:
 - (a) $\{x \in V : x'' + x = 0\}$.
 - (b) $\{x \in V : x'' - 4x' + 3x = 0\}$.
 - (c) $\{x \in V : x''' - 6x'' + 11x' - 6x = 0\}$.
- (16) Show that if U and W are subspace of $\mathbb{R}^9$ such that $\dim U = 5 = \dim W$, then $U \cap W \neq \emptyset$.
- (17) Let $U = \{(a, b, c, d) \in \mathbb{R}^4 : b = -a\}$ and $W = \{(a, b, c, d) : c = -a\}$. Find the dimensions of the subspaces $U, W, U + W$, and $U \cap W$ of $\mathbb{R}^4$.
- (18) Is $\text{span}\{e_1 + e_2, e_2 + e_3, e_3 + e_1\}$ a proper subspace of $\mathbb{R}^3$? Why?
- (19) Prove that the only nonzero proper subspaces of $\mathbb{R}^2$ are the straight lines passing through the origin.
- (20) Show that $U := \{\alpha t^3 + \beta t^7 : (\alpha, \beta) \in \mathbb{R}^2\}$ is a subspace of $\mathcal{P}$. Find a subspace W of $\mathcal{P}$ such that $U \oplus W = \mathcal{P}$.
- (21) In each of the following subspaces U and W of a vector space V , determine the bases and dimensions of $U, W, U \cap W$ and of $U + W$.
 - (a) $V = \mathbb{R}^3$, $U = \text{span}\{(1, 2, 3), (2, 1, 1)\}$, $W = \text{span}\{(1, 0, 1), (3, 0, -1)\}$.
 - (b) $V = \mathbb{R}^4$, $U = \text{span}\{(1, 0, 2, 0), (1, 0, 3, 0)\}$,
 $W = \text{span}\{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 1)\}$.

- (c) $V = \mathbb{C}^4$, $U = \text{span}\{(1, 0, 3, 2), (10, 4, 14, 8), (1, 1, -1, -1)\}$,
 $W = \{(1, 0, 0, 2), (3, 1, 0, 2), (7, 0, 5, 2)\}$.
- (22) Consider each polynomial in $\mathcal{P}$ as a function from the set $\{0, 1, 2\}$ to $\mathbb{R}$. Is the set of vectors $\{t, t^2, t^3, t^4, t^5\}$ linearly independent?
- (23) Given real numbers $a_0, a_1, \dots, a_k$, let V be the set of all solutions $x \in C^k[a, b]$ of the differential equation

$$a_0 \frac{d^k x}{dt^k} + a_1 \frac{d^{k-1} x}{dt^{k-1}} + \dots + a_k x = 0.$$

- Show that V is a vector space over $\mathbb{R}$. What is $\dim(V)$?
- (24) Consider the set S of all vectors in $\mathbb{R}^4$ whose components are either 0 or 1. How many subsets of S are bases for $\mathbb{R}^4$?
- (25) Suppose the vectors $v_1, \dots, v_n$ span the space V . Show that the vectors $v_1, v_2 - v_1, \dots, v_n - v_1$ also span V . Show also that if $v_1, \dots, v_n$ are linearly independent, then $v_1, v_2 - v_1, \dots, v_n - v_1$ are linearly independent.
- (26) Let V be a vector space. Suppose that $v_1, \dots, v_n, \dots$ are in V such that for each $m \in \mathbb{N}$, the vectors $v_1, \dots, v_m$ are linearly independent. Show that $\dim(V) = \infty$.
- (27) Show that $\dim(\mathbb{F}^\infty) = \infty$.
- (28) Show that $\dim(\mathcal{C}[0, 1]) = \infty$.
- (29) Let U and W be subspaces of $\mathbb{F}^7$ with $\dim(U) = 4$ and $\dim(W) = 3$. Show that $U + W = \mathbb{F}^7$ iff $U \cap W = \{0\}$ iff $\mathbb{F}^7 = U \oplus W$.
- (30) Let U be a subspace of a vector space V . Show that $\dim(U) \leq \dim(V)$. Further, if U is a finite dimensional proper subspace of V , then show that $\dim(U) < \dim(V)$.
- (31) Let U and W be subspaces of a vector space of dimension $2n + 1$. Show that if $\dim(U) = \dim(W) \geq n + 1$, then $U \cap W$ contains a nonzero vector.
- (32) Determine the quotient space $\mathcal{P}/U$ for each of the following cases:
- $U = \mathcal{P}_n$.
 - U is the subspace of all polynomials of even degree.
 - U is the subspace of all polynomials which are divisible by t^n for a fixed n .
- (33) Let U and W be subspaces of a finite dimensional vector space V . Prove that any basis of $(U + W)/U$ is in one-one correspondence with any basis of $W/(U \cap W)$.
- (34) Let U be a subspace of a vector space V . Let $u, v, x, y \in V$. Let $\alpha, \beta \in \mathbb{F}$. Show that if $u \equiv_U x$ and $v \equiv_U y$, then $\alpha u + \beta v \equiv_U \alpha x + \beta y$.

1.6. Assignment-Problems: Additional problems.

- (1) Suppose S is a set consisting of n elements and V is the set of all real valued functions defined on S . Show that V is a vector space of dimension n .
- (2) Let U and W be subspaces of a vector space V . Prove the following:
 - (a) $U \cup W = V$ iff $U = V$ or $W = V$.
 - (b) $U \cup W$ is a subspace of V iff $U \subseteq W$ or $W \subseteq U$.
 - (c) Let $U \cap W = \{0\}$. If $v \in U + W$, then there are unique $u \in U, w \in W$ such that $v = u + w$.
- (3) Let V be a vector space. Prove that V is not a finite union of proper subspaces.
- (4) Let $\ell^1(\mathbb{N}) := \left\{x \in \mathcal{F}(\mathbb{N}, \mathbb{F}) : \sum_{j=1}^{\infty} |x(j)| < \infty\right\}$, where $\mathcal{F}(\mathbb{N}, \mathbb{F})$ is the space of all scalar sequences. Show that $\ell^1(\mathbb{N})$ is a subspace of $\mathcal{F}(\mathbb{N}, \mathbb{F})$.
- (5) For a nonempty set Ω , let $\ell^\infty(\Omega) := \{x \in \mathcal{F}(S, \mathbb{F}) : \sup_{s \in S} |x(s)| < \infty\}$. Note that $\ell^\infty(\Omega)$ is the set of all bounded functions on S . Thus, $x \in \ell^\infty(\Omega)$ if and only there exists $M_x > 0$ such that $|x(s)| \leq M_x$ for all $s \in S$. In particular, $\ell^\infty(\mathbb{N})$ is the set of all bounded sequences of scalars.
Show that $\ell^\infty(\Omega)$ is a subspace of $\mathcal{F}(S, \mathbb{F})$
- (6) Show that

$$\begin{aligned} c_{00} &:= \{x \in \mathcal{F}(\mathbb{N}, \mathbb{F}) : \exists k \in \mathbb{N} \text{ such that } x(n) = 0 \forall n \geq k\}, \\ c_0 &:= \{x \in \mathcal{F}(\mathbb{N}, \mathbb{F}) : x(n) \rightarrow 0 \text{ as } n \rightarrow \infty\}, \\ c &:= \{x \in \mathcal{F}(\mathbb{N}, \mathbb{F}) : (x(n)) \text{ converges} \} \end{aligned}$$

are subspaces of $\ell^\infty(\mathbb{N})$. We observe that $c_{00} \subseteq \ell^1(\mathbb{N}) \subseteq c_0 \subseteq c \subseteq \ell^\infty(\mathbb{N})$. Are these inclusions proper?

- (7) Suppose Λ is a set, and for each $\lambda \in \Lambda$ let V_λ be a subspace of a vector space V . Then prove that $\cap_{\lambda \in \Lambda} V_\lambda$ is a subspace of V .
- (8) If one defines $U - W = \{u - v : u \in U, v \in V\}$ for subspaces U, W of a vector space V , then which of the following would hold and which do not?

$$U - U = \{0\}, \quad U - W = U + W, \quad (U - W) + W = U.$$
- (9) Let U, W, X be subspaces of a vector space V . Prove or give a counter-example: if $U \oplus W = U \oplus X$, then $W = X$.
- (10) Show that a subset $\{u_1, \dots, u_n\}$ of V is linearly independent iff the function $(\alpha_1, \dots, \alpha_n) \mapsto \alpha_1 u_1 + \dots + \alpha_n u_n$ from $\mathbb{F}^n$ into V is injective.
- (11) Answer the following questions with justification:
 - (a) Is every subset of a linearly independent set linearly independent?
 - (b) Is every subset of a linearly dependent set linearly dependent?
 - (c) Is every superset of a linearly independent set linearly independent?
 - (d) Is every superset of a linearly dependent set linearly dependent?
 - (e) Is union of two linearly independent sets linearly independent?
 - (f) Is union of two linearly dependent sets linearly dependent?
 - (g) Is intersection of two linearly independent sets linearly independent?
 - (h) Is intersection of two linearly dependent sets linearly dependent?

- (12) Let U be a finite dimensional subspace of a vector space V . Show that any subspace W of V that satisfies $U + W = V$ and $U \cap W = \{0\}$ has dimension $\dim(V) - \dim(U)$.
- (13) Let $U_1, \dots, U_k$ be subspaces of a vector space V . Prove that $V = U_1 \oplus \dots \oplus U_k$ iff for all $x_1 \in U_1, \dots, x_k \in U_k$, $x_1 + \dots + x_k = 0$ implies that $x_1 = \dots = x_k = 0$.
- (14) Let V be an infinite dimensional vector space. Show that there exists a sequence $v_1, v_2, \dots$, of vectors in V such that for each $n \in \mathbb{N}$, the vectors $v_1, \dots, v_n$ are linearly independent.
- (15) For $\lambda \in [a, b]$, let $u_\lambda(t) = \exp(\lambda t)$, $t \in [a, b]$. Show that $\{u_\lambda : \lambda \in [a, b]\}$ is an uncountable linearly independent subset of $\mathcal{C}[a, b]$.
- (16) Given $a_0, a_1, \dots, a_n \in \mathbb{R}$, let

$$V = \{f \in C^k[0, 1] : a_n f^{(n)}(t) + \dots + a_1 f^{(1)}(t) + a_0 f(t) = 0 \forall t \in [0, 1]\}.$$
 Show that V is a subspace of $C^k[0, 1]$, and find its dimension.
- (17) Show that if $\{u_1, \dots, u_n\}$ is a linearly independent subset of a vector space V , and if W is a subspace of V such that $\{u_1, \dots, u_n\} \cap W = \emptyset$, then every x in the span of $\{u_1, \dots, u_n, W\}$ can be written uniquely as $x = \alpha_1 u_1 + \dots + \alpha_n u_n + y$ with $(\alpha_1, \dots, \alpha_n) \in \mathbb{F}^n$, $y \in W$.
- (18) Show that if E_1 and E_2 are linearly independent subsets of V such that $\text{span}(E_1) \cap \text{span}(E_2) = \{0\}$, then $E_1 \cup E_2$ is linearly independent.
- (19) Let $p_1(t), \dots, p_{n+1}(t) \in \mathcal{P}_n$ be such that $p_1(1) = \dots = p_{n+1}(1) = 0$. Show that the polynomials $p_1(t), \dots, p_{n+1}(t)$ are linearly dependent in $\mathcal{P}_n$.
- (20) Let $U_1, \dots, U_n$ be finite dimensional subspaces of a vector space V . Prove that $\dim(U_1 + \dots + U_n) \leq \dim(U_1) + \dots + \dim(U_n)$.
- (21) Let U, W and X be subspaces of a finite dimensional vector space V . Are the following true?
 (a) If $U \oplus W = V$, then $W \oplus U = V$?
 (b) If $U \oplus (W \oplus X) = V$, then $(U \oplus W) \oplus X = V$.
- (22) Let $U_1, \dots, U_n$ be subspaces of a finite dimensional vector space V . Prove the following.
 (a) If $\dim(U_1) + \dots + \dim(U_n) = \dim(V)$, then $V = U_1 \oplus \dots \oplus U_n$.
 (b) $\dim(U_1 \oplus \dots \oplus U_n) = \dim(U_1) + \dots + \dim(U_n)$.
 (c) Each vector in V can be written as a unique sum of vectors from the subspaces $U_1, \dots, U_n$ iff $V = U_1 + \dots + U_n$ and for each $i \in \{2, \dots, n\}$, $U_i \cap (U_1 + \dots + U_{i-1}) = \{0\}$.
 (d) $V = U_1 \oplus \dots \oplus U_n$ iff $V = U_1 + \dots + U_n$ and whenever $0 = u_1 + \dots + u_n$ for $u_1 \in U_1, \dots, u_n \in U_n$, we have $u_1 = \dots = u_n = 0$.
- (23) Let V be a vector space. Show that $\dim(V) = n \geq 1$ iff there exist one dimensional subspaces $U_1, \dots, U_n$ such that $V = U_1 \oplus \dots \oplus U_n$.